

Inverse Laplace Transforms

CLO-II

Chapter#08 (section 8.3)



Section 8.3

Inverse Laplace Transforms

If $F(s)$ is the Laplace transform of a function $f(t)$, i.e. if, $\boxed{\mathcal{L}\{f(t)\} = F(s)}$, then the inverse Laplace transform of $F(s)$, written as $\mathcal{L}^{-1}\{F(s)\}$, is $f(t)$ i.e., $\mathcal{L}^{-1}\{F(s)\} = f(t)$ $\mathcal{L}^{-1}$ (*)

$$\boxed{\mathcal{L}^{-1}\{F(s)\} = f(t)} \quad (1)$$

Thus we see that the inverse Laplace transform recovers the continuous function $f(t)$, when $F(s)$ is given. Therefore, from our knowledge about the Laplace transform we see that,

$$\boxed{\mathcal{L}^{-1}\left\{\frac{1}{s-k}\right\} = e^{kt}} \quad \boxed{\text{ox}} \quad (1)$$

$$\boxed{\frac{1}{s-k} = \mathcal{L}[e^{kt}]} \quad (2)$$

$$\boxed{\mathcal{L}^{-1}\left\{\frac{k}{s^2 + k^2}\right\} = \sin kt} \quad (3)$$

$$\boxed{\frac{k}{s^2 + k^2} = \mathcal{L}[\sin kt]} \quad \boxed{\text{ox}} \quad (4)$$



Example 101

Determine $\mathcal{L}^{-1} \left\{ \frac{3}{s^2 + 35} \right\}$ — (1)

Solution

$$\mathcal{L}^{-1} \left\{ \frac{3}{s^2 + 35} \right\} = \frac{3}{\sqrt{35}} \mathcal{L}^{-1} \left\{ \frac{\sqrt{35}}{s^2 + (\sqrt{35})^2} \right\} \text{ — (2)}$$

$$= \frac{3}{\sqrt{35}} \sin(\sqrt{35} t) \text{ — (3) } (K = \sqrt{35})$$

Rough:

$$\text{As, } \mathcal{L}[\sin kt] = \frac{k}{s^2 + k^2} \quad \text{— (4)}$$

$$\mathcal{L}[\sin \sqrt{35} t] = \frac{\sqrt{35}}{s^2 + (\sqrt{35})^2} \quad \text{— (5)}$$



Example 02

Find $\mathcal{L}^{-1} \left\{ \frac{-3s}{s^2 + 94} + \frac{12}{s-3} \right\}$ — (1)

Solution

$$\begin{aligned} \mathcal{L}^{-1} \left\{ \frac{-3s}{s^2 + 94} + \frac{12}{s-3} \right\} &= -3 \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + (\sqrt{94})^2} \right\} + 12 \mathcal{L}^{-1} \left\{ \frac{1}{s-3} \right\} \\ &= -3 \cos(\sqrt{94}t) + 12 e^{3t} \end{aligned}$$

(2)
(3)

Rough

1) $\boxed{L[\cos kt] = \frac{s}{s^2 + k^2}}$ (4)

$\boxed{L[\cos \sqrt{94}t] = \frac{s}{s^2 + (\sqrt{94})^2}}$
 $\therefore k = \sqrt{94}$ (5)

2) $\boxed{L[e^{at}] = \frac{1}{s-a}}$ (6)

$\boxed{L[e^{3t}] = \frac{1}{s-3}}$
 $\therefore a = 3$ (7)



Example 03

Determine $\mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 6s + 15} \right\}$ — (1)

Solution

$$\mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 6s + 15} \right\} = \mathcal{L}^{-1} \left\{ \frac{s}{(s+3)^2 + 6} \right\} \text{ — (2)}$$

$$= \mathcal{L}^{-1} \left\{ \frac{(s+3)-3}{(s+3)^2 + 6} \right\} = \mathcal{L}^{-1} \left\{ \frac{(s+3)}{(s+3)^2 + 6} - \frac{3}{(s+3)^2 + 6} \right\} \text{ — (4)}$$

L (3)

$$= \mathcal{L}^{-1} \left\{ \frac{(s+3)}{(s+3)^2 + (\sqrt{6})^2} - \frac{3}{\sqrt{6}} \frac{\sqrt{6}}{(s+3)^2 + (\sqrt{6})^2} \right\} \text{ — (5)}$$

$$= \mathcal{L}^{-1} \left[\frac{s+3}{(s+3)^2 + (\sqrt{6})^2} \right] - \frac{3}{\sqrt{6}} \mathcal{L}^{-1} \left[\frac{\sqrt{6}}{(s+3)^2 + (\sqrt{6})^2} \right]$$

L (6)

$$= e^{-3t} \cos \sqrt{6} t - \sqrt{\frac{3}{2}} e^{-3t} \sin \sqrt{6} t$$

L (7)

[Rough]

$$1) \boxed{s^2 + 6s + 15 = s^2 + 6s + 9 + 5} \text{ — (8)}$$

$= (s+3)^2 + 5$

$$2) \boxed{\mathcal{L}[\sin kt] = \frac{k}{s^2 + k^2}, \mathcal{L}[\cos kt] = \frac{s}{s^2 + k^2}} \text{ — (9)}$$

$$3) \boxed{\begin{aligned} \mathcal{L}[e^{at} f(t)] &= F(s-a) \\ \mathcal{L}[e^{-at} f(t)] &= F(s+a) \end{aligned}} \text{ — (10)}$$



Example 04

Evaluate $\mathcal{L}^{-1} \left\{ \frac{1}{(s-5)^3} + \frac{1}{s^2 + 2s - 8} \right\}$ (1)

Solution

$$\begin{aligned} \mathcal{L}^{-1} \left\{ \frac{1}{(s-5)^3} + \frac{1}{s^2 + 2s - 8} \right\} &= \mathcal{L}^{-1} \left\{ \frac{1}{(s-5)^3} \right\} + \mathcal{L}^{-1} \left\{ \frac{1}{(s+1)^2 - 9} \right\} \quad (2) \\ &= \frac{1}{2!} \mathcal{L}^{-1} \left\{ \frac{2!}{(s-5)^3} \right\} + \frac{1}{3} \mathcal{L}^{-1} \left\{ \frac{3}{(s+1)^2 - (3)^2} \right\} \quad (3) \\ &= \frac{1}{2!} \mathcal{L}^{-1} \left\{ \frac{2!}{s^3} \right\}_{s \rightarrow s-5} + \frac{1}{3} \mathcal{L}^{-1} \left\{ \frac{3}{s^2 - (3)^2} \right\}_{s \rightarrow s+1} \quad (4) \\ &= \frac{1}{2} e^{5t} t^2 + \frac{1}{3} e^{-t} \sinh 3t \end{aligned}$$

Rough:

01) $\mathcal{L}[t^2] = \frac{2!}{s^3}$ (6)

$\mathcal{L}[\sinh \omega t] = \frac{\omega}{s^2 - \omega^2}$ (7)

02) $\mathcal{L}[e^{at} f(t)] = F(s-a)$
 $\mathcal{L}[e^{-at} f(t)] = F(s+a)$ (8)

03) $s^2 + 2s - 8$
 $= s^2 + 2s + 1 - 9$
 $= (s+1)^2 - (3)^2$ (9)



Example 05

Evaluate $\mathcal{L}^{-1}\left\{\frac{1}{(s-1)(s+2)(s+4)}\right\}$ — (1)

Solution

In order to evaluate the given inverse Laplace transform, we resolve the function of s into partial fractions. Thus we let,

$$\frac{1}{(s-1)(s+2)(s+4)} = \frac{A}{s-1} + \frac{B}{s+2} + \frac{C}{s+4} \quad \text{--- (2)}$$

e.,

$$1 = A(s+2)(s+4) + B(s-1)(s+4) + C(s-1)(s+2). \quad \text{--- (3)}$$

to find the value of A , we put $s-1 = 0$, or $s = 1$ in the Eqn. (3) to obtain, $1 = A(3)(5)$

e.,

$$A = \frac{1}{15} \quad \text{--- (4)}$$

Again, to find the value of B , we put $s+2 = 0$, or $s = -2$ in the Eqn. (3) to obtain,

$$1 = B(-3)(2), \text{ i.e.,}$$

$$B = -\frac{1}{6} \quad \text{--- (5)}$$



Similarly to evaluate C , we put $s + 4 = 0$, or $s = -4$ in the Eqn. (103). Thus we get,

$$1 = C(-5)(-2), \text{ i.e.,}$$

$$C = \frac{1}{10}.$$

Thus we can write, by putting (4) to (6) in (2).

$$\frac{1}{(s-1)(s+2)(s+4)} = \frac{1/15}{s-1} - \frac{1/6}{s+2} + \frac{1/10}{s+4}$$

Therefore,

$$\mathcal{L}^{-1} \left\{ \frac{1}{(s-1)(s+2)(s+4)} \right\} = \frac{1}{15} \mathcal{L}^{-1} \left\{ \frac{1}{s-1} \right\} - \frac{1}{6} \mathcal{L}^{-1} \left\{ \frac{1}{s+2} \right\} + \frac{1}{10} \mathcal{L}^{-1} \left\{ \frac{1}{s+4} \right\}$$

$$= \frac{1}{15} e^t - \frac{1}{6} e^{-2t} + \frac{1}{10} e^{-4t}$$

Rough:

$$\therefore \mathcal{L}[e^{at}] = \frac{1}{s-a}$$

$$e^{at} = \mathcal{L}^{-1} \left[\frac{1}{s-a} \right]$$



Example 06.

Evaluate $\mathcal{L}^{-1} \left\{ \frac{s+1}{s^2+2s} \right\}$ — (1)

Solution

Here we can make use of the partial fractions to simplify the function of s . We let,

$$\frac{s+1}{s(s+2)} = \frac{A}{s} + \frac{B}{s+2} \quad \text{--- (2)}$$

Thus,

$$s+1 = A(s+2) + Bs = (A+B)s + 2A \quad \text{--- (3)}$$

It gives, by comparing coeff. of s^1, s^0 on both side

$$A+B=1, \text{ and } 2A=1 \quad \text{--- (4)}$$

i.e.,

$$A = \frac{1}{2} = B \quad \text{--- (5)}$$

Therefore,

$$\mathcal{L}^{-1} \left[\frac{s+1}{s^2+2s} \right] = \mathcal{L}^{-1} \left[\frac{1/2}{s} \right] + \mathcal{L}^{-1} \left[\frac{1/2}{s+2} \right] \quad \text{--- (6)}$$

$$= \frac{1}{2} \mathcal{L}^{-1} \left[\frac{1}{s} \right] + \frac{1}{2} \mathcal{L}^{-1} \left[\frac{1}{s+2} \right] \quad \text{--- (7)}$$

$$= \frac{1}{2} + \frac{1}{2} e^{-2t} \quad \text{--- (8)}$$

Rough

$$\mathcal{L}[e^{at}] = \frac{1}{s+a}$$

$$\mathcal{L}[1] = \frac{1}{s}$$

$$\therefore \mathcal{L}^{-1} \left[\frac{1}{s+2} \right]$$

$$1 = \mathcal{L}^{-1} \left[\frac{1}{s} \right]$$

Example 07

Find $\mathcal{L}^{-1} \left\{ \frac{a^2}{s(s+a)^2} \right\}$ — (1)

Solution

We let,

$$\frac{a^2}{s(s+a)^2} = \frac{A}{s} + \frac{B}{s+a} + \frac{C}{(s+a)^2}, \quad \text{— (2)}$$

We find that, $A = 1$, $B = -1$, $C = -a$. Therefore,

$$\mathcal{L}^{-1} \left\{ \frac{a^2}{s(s+a)^2} \right\} = \mathcal{L}^{-1} \left\{ \frac{1}{s} \right\} - \mathcal{L}^{-1} \left\{ \frac{1}{s+a} \right\} + \mathcal{L}^{-1} \left\{ \frac{-a}{(s+a)^2} \right\} \quad \text{— (3)}$$

$$= 1 - e^{-at} - a t e^{-at}. \quad \text{— (4)}$$

Rough

2) $\mathcal{L}^{-1} \left[\frac{1}{s} \right] = 1$

02. $\mathcal{L}^{-1} \left[\frac{1}{s+a} \right] = e^{-at}$

03. $\mathcal{L}^{-1} \left[\frac{1}{(s+a)^2} \right] = t e^{-at}$



EXERCISE - 8.3

Note: Answers are given against the respective questions within square brackets.

Find $f(t)$ if $F(s) = \mathcal{L}\{f(t)\}$ is as under

1. $\frac{1}{s^3}$; $\left[\frac{t^2}{2}\right]$

2. $\frac{1}{s^2} - \frac{48}{s^5}$; $[t - 2t^4]$

3. $\frac{(s+1)^3}{s^4}$; $\left[1 + 3t + \frac{3}{2}t^2 + \frac{1}{6}t^3\right]$

4. $\frac{(s+2)^2}{s^3}$; $[1 + 4t + 2t^2]$

5. $\frac{s-9}{s^2-9}$; $[\cosh 3t - 3 \sinh 3t]$

6. $\frac{6}{s^2+49}$; $\left[\frac{6}{7} \sin 7t\right]$

7. $\frac{16s}{s^2+16}$; $[16 \cos 4t]$

8. $\frac{4s}{4s^2+1}$; $\left[\cos \frac{t}{2}\right]$

9. $\frac{1}{s^2-16}$; $\left[\frac{1}{4} \sinh 4t\right]$

10. $\frac{2s-6}{s^2+9}$; $[2 \cos 3t - 2 \sin 3t]$

11. $\frac{s+1}{s^2+2}$; $\left[\cos \sqrt{2} t + \frac{1}{\sqrt{2}} \sin \sqrt{2} t\right]$

12. $\frac{1}{s^2+3s}$; $\left[\frac{1}{3}(1 - e^{-3t})\right]$

13. $\frac{s}{s^2+2s-3}$; $\left[\frac{1}{4}(3e^{-3t} + e^t)\right]$

